IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

Wednesday 10 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the feedback system in Figure 1. Here, $G(s)$ is a nominal system model and $K(s)$ is a controller. The transfer matrix $\Delta_m(s)$ denotes a stable uncertainty. The design specifications are as follows:

   i) The robust stability specification is to design a controller $K(s)$ such that the feedback loop is internally stable when $\|\Delta_m(j\omega)\| \leq |w_m(j\omega)|, \forall \omega$, where $w_m(s)$ is an appropriate weighting function.

   ii) The performance specifications are to design an internally stabilizing controller such that when $\Delta_m = 0$, the closed-loop should satisfy:

$$\|u(j\omega)\| < |w_u(j\omega)^{-1}| \|r(j\omega)\|,$$

   and

$$\|y(j\omega)\| < |w_y(j\omega)^{-1}| \|r(j\omega)\|,$$

   for all ω, where $w_u(s)$ and $w_y(s)$ are appropriate weighting functions.

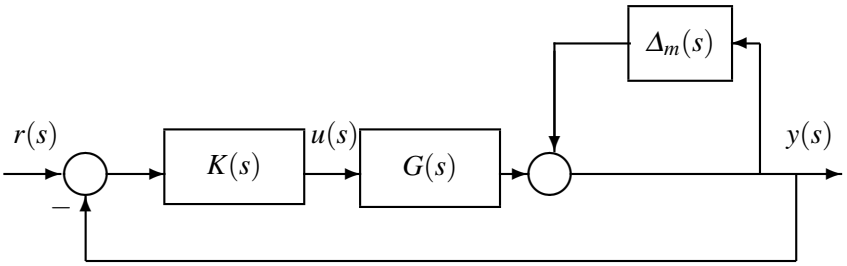


Figure 1

   a) State the small gain theorem concerning the internal stability of a feedback loop with a forward transfer matrix Δ and a feedback transfer matrix H. [5]

   b) Use the small gain theorem to derive $\mathscr{H}_\infty$–norm bounds, in terms of $G(s)$, $K(s)$, $w_m(s)$, $w_u(s)$ and $w_y(s)$ that are sufficient to achieve the specifications. [5]

   c) Define appropriate cost, external, control and measured signals and draw a block diagram, showing these signals, the nominal system model, the controller, and weighting functions. [5]

   d) Derive and draw a generalised regulator problem that captures these sufficient conditions. [5]

2. Consider the linear quadratic regulator (LQR) in Figure 2 for which $x(0) = x_0$, where $d(t)$ denotes a disturbance and $\alpha > 0$ is given. Let $z = \begin{bmatrix} z_1^T & z_2^T \end{bmatrix}^T$. A stabilizing state-feedback gain matrix F is to be designed such that, when $d(t) = 0, \forall t$, the cost function $J = \|z\|_2^2$ is minimized.

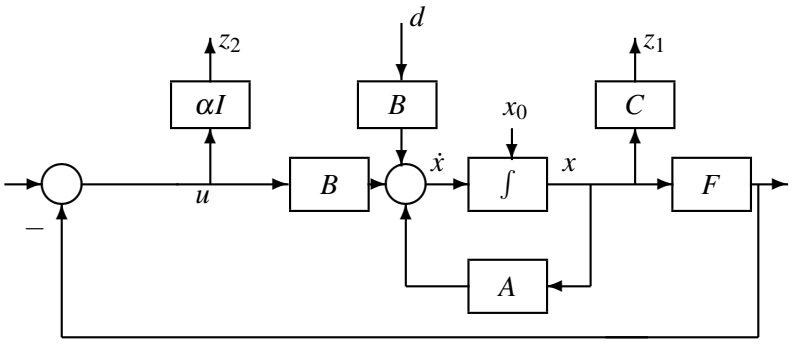


Figure 2

a) Define a suitable Lyapunov function $V(t)$ and obtain an expression for $\int_0^\infty \dot{V}(t)dt$. State clearly any assumptions made. [5]

b) Find an expression for F that minimizes J. Give also the algebraic Riccati equation that needs to be satisfied. What is the minimum value of J? [5]

c) Prove now that the closed-loop system in Figure 2 is stable. State the assumptions required to guarantee stability. [5]

d) By evaluating a state–space realisation for $T_{z_1 d}(s)$ in Figure 2 and using the Bounded Real Lemma, derive an upper bound on $\|T_{z_1 d}\|_\infty$. Ignore issues relating to strict matrix inequalities. [5]

3. Consider the $\mathscr{H}_\infty$ regulator given by

$$\begin{bmatrix} z(s) \\ y(s) \end{bmatrix} = P(s) \begin{bmatrix} w(s) \\ u(s) \end{bmatrix}, \; u(s) = Fy(s), P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \stackrel{s}{=} \left[\begin{array}{c|c|c} A & B & B \\ \hline C & 0 & 0 \\ 0 & 0 & I \\ \hline I & 0 & 0 \end{array}\right],$$

for which it is assumed that the triple (A,B,C) is minimal and the state of the system is initially at rest. Let $H(s)$ denote the transfer matrix from w to $z = [\; z_1^T \;\; z_2^T \;]^T$ and let $\gamma > 0$ be given.

A stabilizing state-feedback gain matrix F is to be designed such that $\|H\|_\infty \leq \gamma$.

a) Write down a block diagram representing the feedback loop for the generalised regulator. Clearly indicate all the signals. [5]

b) By defining a suitable Lyapunov function and a cost function J involving an auxiliary variable $X = X'$, and carrying out two completions of squares, derive an expression for J that can be used to solve the design problem. [5]

c) Use the expression for J to formulate the design problem and derive sufficient conditions for its solution. Your conditions should be in the form of the existence of a certain solution to an algebraic Riccati equation. It should also include an expression for F and an expression for the worst–case disturbance w^*. [5]

d) Let $\gamma = 1$. Write down the Riccati equation and give a condition for the existence of a stabilizing solution. [5]

4. a) Consider a state–variable model described by the dynamics

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad y(t) = Cx(t) + Du(t),$$

and let $H(s) = D + C(sI - A)^{-1}B$ denote the corresponding transfer matrix. Assume that A is stable and let $\gamma > 0$ be given. Derive sufficient conditions for $\|H\|_\infty < \gamma$ in the form of matrix inequalities involving a Lyapunov matrix P and the system matrices A, B, C and D. [5]

b) Consider the $\mathscr{H}_\infty$ filter for estimating Fx, for a given F, shown in Figure 4.

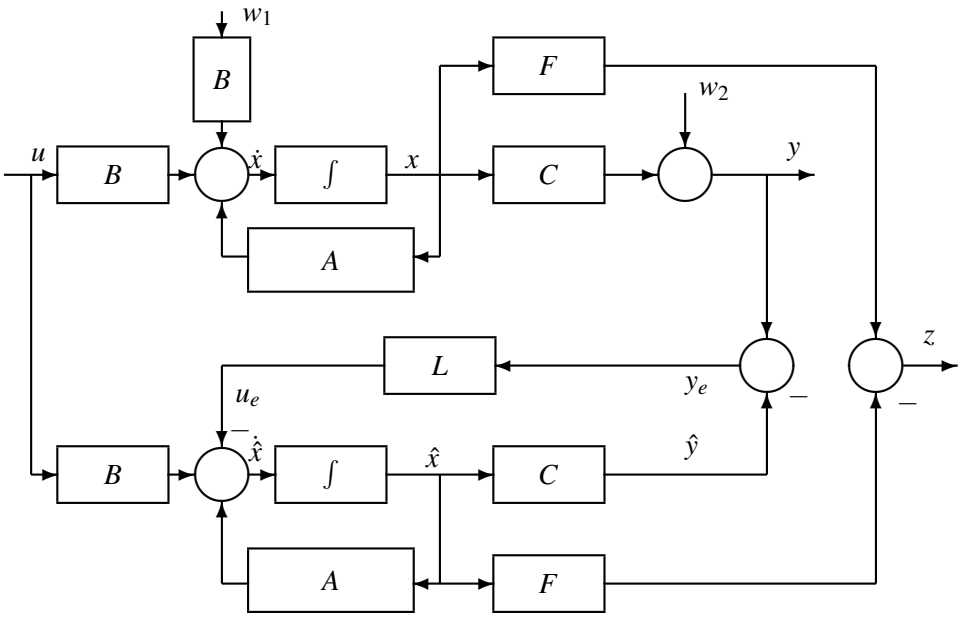


Figure 4

Let $T_{zw}(s)$ denote the transfer matrix from $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ to z. A stabilizing filter gain matrix L is to be designed such that, for $\gamma > 0$, $\|T_{zw}\|_\infty < \gamma$.

i) Derive a state-space realisation for the transfer matrix $T_{zw}(s)$. [5]

ii) Use the Bounded Real Lemma, derived in Question 4a above, to derive sufficient conditions for $\|T_{zw}\|_\infty < \gamma$. These conditions should be in the form of matrix inequality constraints. [5]

iii) Suggest a transformation of variables that turns the matrix inequality derived in Part (ii) above into a linear matrix inequality. [5]